PERTRONIC
ISOLATE TIMER MODULE
AUSTRALIA
TECHNICAL MANUAL

Valid For:

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Issue 2.4

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Acronyms:

ITM	Isolate Timer Modules

Glossary:

F100A	Pertronic F100A series Fire Alarm Panel
F120A	Pertronic F120A series Fire Alarm Panel
F220	Pertronic F200 series Fire Alarm Panel

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1 INTRODUCTION

1.1 Functional Description

The Pertronic Isolate Timer Module (ITM) provides a Countdown Timer for isolating a zone for a pre-determined period. It may be used to isolate devices in a particular zone, when work that is being performed may cause an alarm.

The Pertronic ITM:

- Allows a zone to be isolated at the push of a button.
- Provides the ability to isolate a zone for a user defined period of time.
- Provides two separate clean-contact changeover connections for external equipment.
- Both clean contact outputs are active during the Isolate condition.
- Provides visual indication that the zone has been isolated by the panel.
- Provides Visual and Audible warning that the isolated countdown period is about to expire.
- Connects to and is powered from Pertronics analogue addressable loop
- Can be used to isolate VESDA equipment connected to Pertronic Fire Alarm Panels, or 3rd Party products.
- May be mounted in standard single-gang flush or surface-mount electrical fittings and is supplied with a protective plastic cover for installation during building construction.

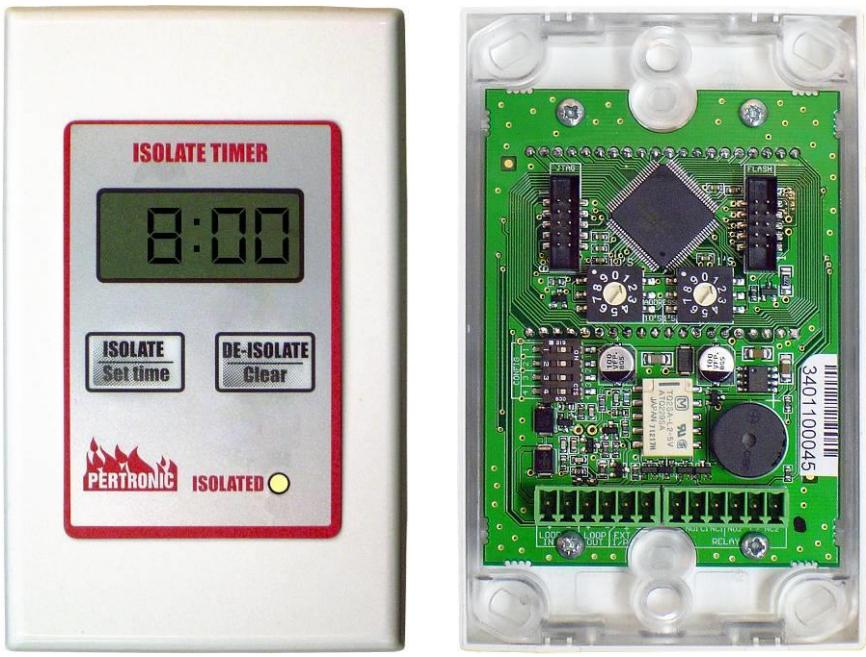

Figure 1.1: Isolate Timer Module (ITM)

1.2 Features

Basic ITM operation is summarised below

- The Isolate Button increments the countdown timer (maximum 8 hours) and sends the Isolate command to the Panel
- The De-Isolate Button clears the countdown timer and sends the De-isolate command to the Panel. If the zone is in Alarm, a warning alerts if the isolated zone is in alarm
- The Isolate LED provides real time indication that the zone has been isolated by the panel
- An LCD Display provides the real-time status of the Panel. If a device is in 'Alarm' and the countdown time is about to expire, the user may increase the isolate time to prevent a false activation.
- In the Last 5 minutes of countdown, an on-board Buzzer beeps once every 10 seconds. If the ITM detects the presence of alarm in the zone in the last one minute, the ITM gives a very agitated signal to warn the user of an impending brigade call. Audible indication can be silenced by either extending or clearing the countdown time.
- At any time during the countdown, if the user attempts to de-isolate the zone with an alarm present, the buzzer beeps agitated. Only at the end of the 30 second period does the de-isolation occur. This enables the user to punch in extra time if desired & clear the system before proceeding with the de-isolation.
- In analogue addressable mode the External input allows device operation to be locked by an external key-switch. In Standalone mode, the external input may be used for Alarm feedback indication from 3rd party products.
- Uses two analogue addresses and is easily configured using Pertronic configuration software
- Each unloaded Zone Isolate Timer draws 15mA in the quiescent state
- Integrated within a Series 500 PDL flush plate, allowing easy mounting. A range of switch covers may be used to suit any décor.
- Compatible with Pertronic Analogue Fire Control Panels with the following firmware:

Panel:	F100A-128	v4.56 or greater
	F100A-256	v5.xx
	F120A	v4.02A or greater
	F220	
Loop Driver:	F120 (old) Loop Driver	v1.38 or greater
	F120 Loop Driver II	v2.12
Utilities:	F100 Config Utility	v3.18.01 or greater
	F120 Config utility	v2.28 or greater
	FireUtils	

For panel configuration details, please refer Section 3 Fire Panel Configuration.

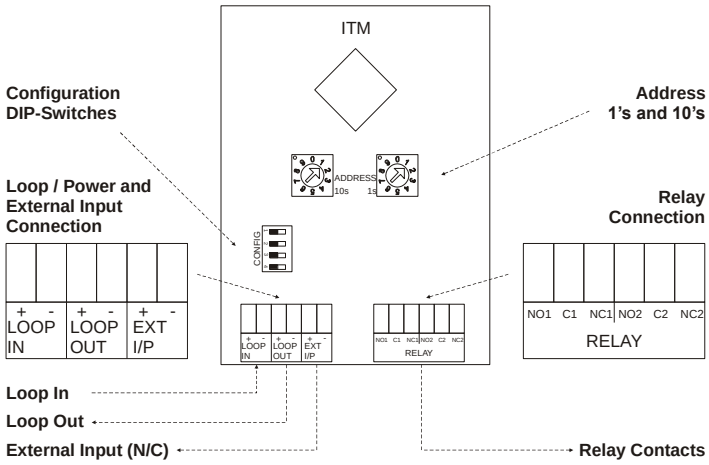

Figure 1.2: Isolate Timer Module Block Diagram

1.3 Specification

Dimensions: 117 x 74 x 12 (height above flush box). H x W x Dmm

- designed to mount in a standard electrical flush box

Loop Input/Output: Used for connection to the analogue loop, or for supplying power in 'stand-alone' mode

Inactive current: 24V, 15mA (i.e. de-isolate state)

Active current: 24V, 18mA (i.e. isolate state)

> **Note:** The maximum total loop current allowed for all connected loop devices is 350mA per loop. Use this current limit to determine the number of ITM's that can be installed on each loop. The loop voltage must not fall below 18Vdc under 'Alarm' conditions.

Addressing: In analogue addressable mode, each ITM uses two loop addresses. The Address Switches select the base or start address of the ITM.

Address 'n': Input configured as ISO

Address 'n+1': Output configured as RELAY

Button Operation: Two buttons provide simple operation of the ITM:

- Isolate/Set Time Button: Sets the countdown timer (Min 2 minutes, max 8 hours) and sends the Isolate command to the Panel. Refer to Table 1-1: ITM Countdown times.
- De-Isolate/Clear Button: If zone is 'Normal', pressing the button clears the countdown timer and sends the de-isolate command to the Panel. If zone is in 'Alarm', an audible warning alerts sounds, and countdown time is set to 30 seconds.

Depending on the mode of operations, the keys may be disabled by using the keyswitch input. Refer to Dip Switch Settings.

Isolate LED: A yellow Isolate LED provides 'real time' indication that the zone has been isolated. In 'Analogue' mode, the Isolate LED is controlled from the panel. In 'Stand-alone' mode, the Isolate LED is controlled by the ITM.

Alarm Indication: When a detector inthe same zonee as the ITM is in 'Alarm', an 'A' is displayed on the LCD. In 'Analogue' mode, alarm indication is controlled from the panel. In 'Stand-alone' mode, the 'Alarm' indication is determined by the state of the 'Ext Input' input.

> **Note:** The ITM must **NOT** be used to isolate previously activated smoke alarms. It is recommended that only trained personal operate the ITM. In areas where the ITM is accessible by civilians, the ITM must be secured by the use of a key switch.

Relay Operation: The Relay energises when the 'Isolate" LED in on. Two sets of 24V 2A clean contacts are provided.

LCD Indication:

The LCD displays the following information:

- Time before Isolation period expires. Displayed in hours and minutes, except in the last minute of countdown where it changes to seconds.
- Zone 'Alarm' state. An 'A' is displayed when a device in the zone is in 'Alarm'.
- Defect / Fault information. Refer to Table 1-3: ITM Defect Messages

Backlight:

The backlight turns on in response to any keypress, and turns off 10 seconds after the last keypress. During the last 5 minutes of countdown, the backlight remains on.

Ext Input:

Can be configured in two modes of operation.

- Used to secure the ITM from unintended use by disabling the keypad. May be connected to an external switch or clean contact relay. Normally shorted, open circuit to disable keypad.
- Alarm activation indication during 'Stand-Alone' mode. Normally shorted, open circuit to indicate alarm.

Dip Switch:

Used to select the ITM's mode of operation.

- SW1-1 Selects operating mode:
  OFF: ITM operates in analogue addressable mode
  ON: ITM operates in standalone mode.
- SW1-2 Selects the mode in which the Ext Input operates during stand- alone mode only.
  OFF: Ext Input used to lock keypad. Normally closed to operate ITM, open to lock buttons.
  ON: Ext input used for Alarm feedback indication. An open circuit on input displays a 'A' on the LCD.
- SW1-3 Selects country mode:
  OFF: New Zealand
  ON: Australia
- SW1-4 Used in factory only:
  OFF: Normal
  ON: Factory test

Buzzer:

Depending on the state of the ITM, the buzzer can sound a number of cadences. The buzzer can be silenced at any time by extending or clearing the countdown time. Refer to Table 1-2: ITM Buzzer cadences

Defect Indication:

A defect message 'dEF' followed by a number, is displayed on the LCD when either a loop, or key defect occurs. Refer to Table 1-3: ITM Defect Messages.
NOTE: If an ITM address is not configured in the Panel, the Panel will signal a defect on the configured address.

Note:
The Isolate Timer Module uses the microcontroller's internal RC oscillator to generate the time delay. This oscillator has a theoretical worst case accuracy of ±10%. Typical measured accuracy is less than ±2%

Number of Isolate Key Presses	Description
1st	2 minutes
2nd	5 minutes
3rd	30 minutes
4th	1 hour
6th	2 hours
7th	3 hours
8th	4 hours
9th	5 hours
10th	6 hours
11th	7 hours
12th	8 hours

Table 1-1: ITM Countdown times.

Buzzer Cadence	Description
Single beep	Button has been pressed
Four 100ms beeps in quick succession	Button has been pressed while ITM is in secure state.
One beep every 10 seconds	The ITM has been activated, and there is only five minutes left on the clock before isolate deactivation.
Continuous 50ms beeps	The ITM has been activated, a device within the same zone of the ITM is in alarm; and there is one minute left on the clock before isolate deactivation.
Continuous	The ITM is less than 10 seconds from isolate de-isolation

Table 1-2: ITM Buzzer cadences

Defect Message	Description
dEF 1	Address 'n' loop communication, has been lost for more than one minute*
dEF 2	Address 'n+1' loop communication has been lost for more than one minute*
dEF 3	Either Isolate or De-isolate key has been pressed and held for more than 15 seconds.

Table 1-3: ITM Defect Messages

2 SYSTEM FEATURES

2.1 Analogue or Standalone Mode Selection

The Isolate Timer Module can operate in one of two modes, determined by whether the ITM is connected to a Pertronic Analogue loop e.g. F100 and F120, or to a third party system.

2.1.1 Analogue Mode

Analogue Mode is used when the ITM is connected to either a Pertronic F100 or F120 Fire Alarm Panel. Power, Isolate and Alarm indication are all provided via the analogue addressable loop.

A typical installation is shown in Figure 2.1

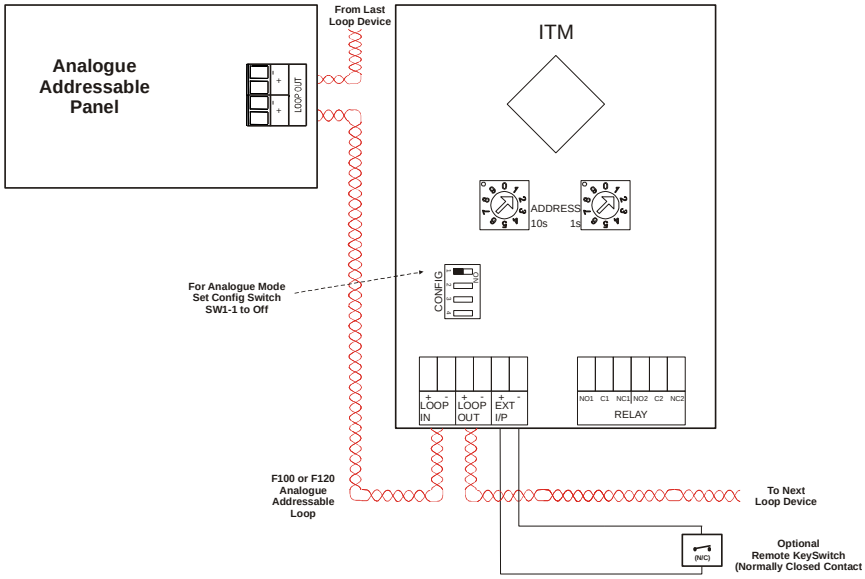

Figure 2.1: Typical Analogue Mode Installation

Analogue mode is selected by setting Dipswitch SW1-1 to OFF. In this mode, address ‘n’ on the panel is set as ‘ISO SWITCH’, and address ‘n+1’ is set as ‘RELAY’.

2.1.1.1 Operation.

When the Isolate button is pressed, the ITM:

- Changes to Isolate state
- Signals a ‘Isolate command’ to the Fire Control Panel
- When an ‘Isolate command is received from the panel, energizes the on-board Isolate Relay and illuminates the Isolate LED.
- Displays an ‘A’ on the LCD, when an ‘Alarm’ command is received from the panel

When the De-Isolate button is pressed, or the countdown time expires, the ITM:

- Stops sending the ‘Isolate command’ to the Fire control panel.
- De-energizes the on-board Isolate relay, and turns off the ‘Isolate’ LED when the ‘Isolate’ command is no longer received from the panel
- Reverts to the standby state

While in standby state, the ITM:

- Defaults the LCD to flash the colon every second
- Displays an ‘A’ on the LCD, when an ‘Alarm’ command is received from the panel

2.1.1.2 Addressing

Select the base address (01 to 98) using the two loop address switches labeled ADDRESS located on the back of the PCB.

The ITM uses two module loop addresses - the first or base address for the Isolate input, the second address (base address 'n' + 1) for Alarm Indication.

Since, each loop can have a maximum of 99 modules (addresses 01 to 99), so the highest base address the ITM can have is 98.

The maximum number of ITMs, which can be connected on each loop is limited by the current the loop can supply (350mA). Each Isolate Timer Module draws 15mA in the quiescent state, 18mA in Active state

When multiple ITM's (or any devices configured as an 'Isol' type), are configured within a zone, Zonal 'Isolate' indication will be displayed on each ITM. However 'Alarm' indication will only occur on ITMs that have entered the countdown state.

Isolation Precedence

If a zone contains any devices that are currently isolated **then these devices will also be de-isolated** when the ITM de-isolates the zone.

If the previously isolated device is in an **ALARM** state then immediately the zone is de-isolated the fire panel will signal an **ALARM**.

It is recommended that any device isolations for zones controlled by ITM's should only be made by service personnel and remain only for the duration that personnel are available on-site.

2.1.2 Stand Alone Mode

Stand Alone Mode is used when the ITM is required to interface 3rd Party Products. In this mode, the Loop Input connection is connected to a power supply and is only used for supplying power to the device. Loop communication defects on ITM are suppressed.

A typical installation is shown in Figure 2.1

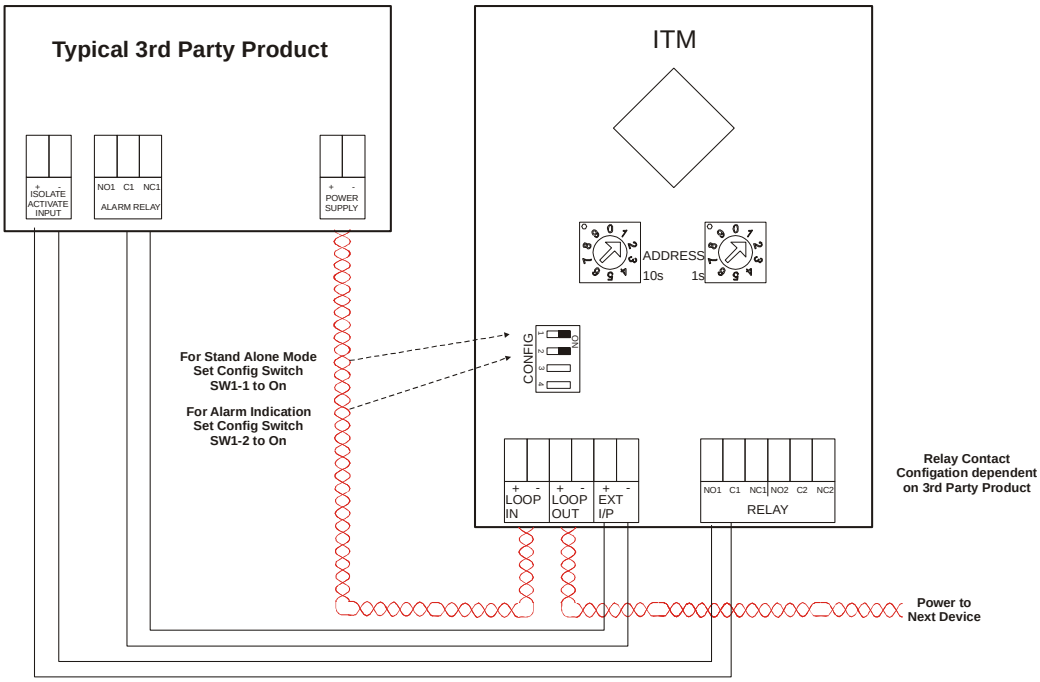

Figure 2.2: Typical Stand Alone Mode Installation

Stand Alone mode is selected by setting Dipswitch SW1-1 to ON. External Alarm Indication is selected by setting Dipswitch SW1-2 to ON, and using the Ext Input for Alarm feedback information. The Ext Input must only be connected to normally closed clean contacts.

2.1.2.1 Operation.

When the Isolate button is pressed, the ITM:

- changes to the Isolate state
- energizes the on-board Isolate Relay
- Illuminates the 'Isolate' LED
- Displays an 'A' on the LCD, when the Ext input is configured as an 'Alarm' input

When the De-Isolate button is pressed, or the countdown time expires, the ITM:

- de-energizes the on-board Isolate Relay
- Turns off the 'Isolate' LED
- reverts to standby state

While in standby mode, the ITM:

- Defaults the LCD to flash the colon every second
- Displays an 'A' on the LCD, when the Ext Input is configured as an 'Alarm' input

3 FIRE PANEL CONFIGURATION

The F100A, F120A and F220 panels require specific configuration setups for the ITM to operate correctly:

1. The device at the ITM's base address must have the type ISO and have its zone number set to the zone that requires isolating. No more than one zone can be isolated by the ITM.
2. The device at the ITM's second address (base address +1) must have the type RLY and have its Zone number set to 0.

These apply to all the above panels. The specific configuration steps for each panel are given below.

3.1 F100A-128 Analogue Addressable Panel (s/w 4.xx)

3.1.1 Isolate Timer Module Configuration:

Key: Y = Yes N = No o = optional z = zone to be isolated	**Address**	**Type**	**Zone**	**Sens (Day)**	**Sens (Night)**	**Latching**	**AVF**	**Warn. Sys.**	**Ext Bell**	**ACF**	**Door**	**Fire**	**GP Relay**	**Fire Queue**	**Visible**	**Logged**	**Buzzer**	**Brigade Latch**	**Output 1**	**Output 2 - 4**
	n	ISO	z			N	N	N	N	N	N	N	N	N	Y	Y	N	N	ITMn+1	o
	n+1	RLY	0																	

Table 3-1: Isolate Timer Module F100-128 Configuration

The 'Visible' and 'Logged' flags for the ITM isolate switch device address should be set to YES; all other flags remain un-configurable. Map an output from the ITM's ISO device to the ITM RLY device. This will activate the Alarm Indication on the ITM LCD by displaying 'A' (for alarm) when the isolated zone is in alarm.

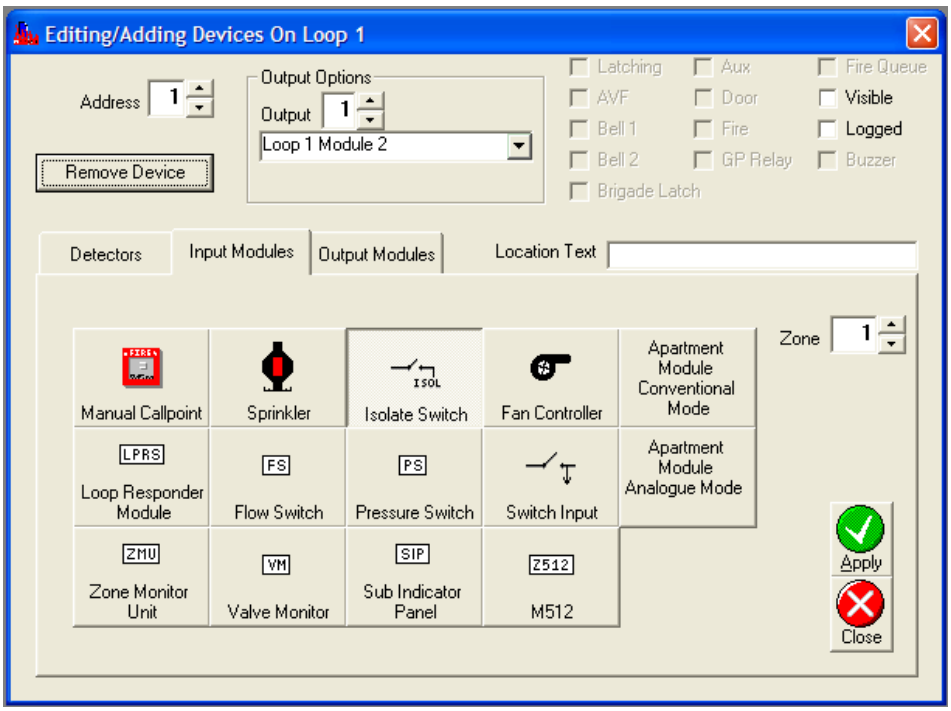

Figure 3.1: F100-128 Isolate Timer Module
Address 'n' = 1

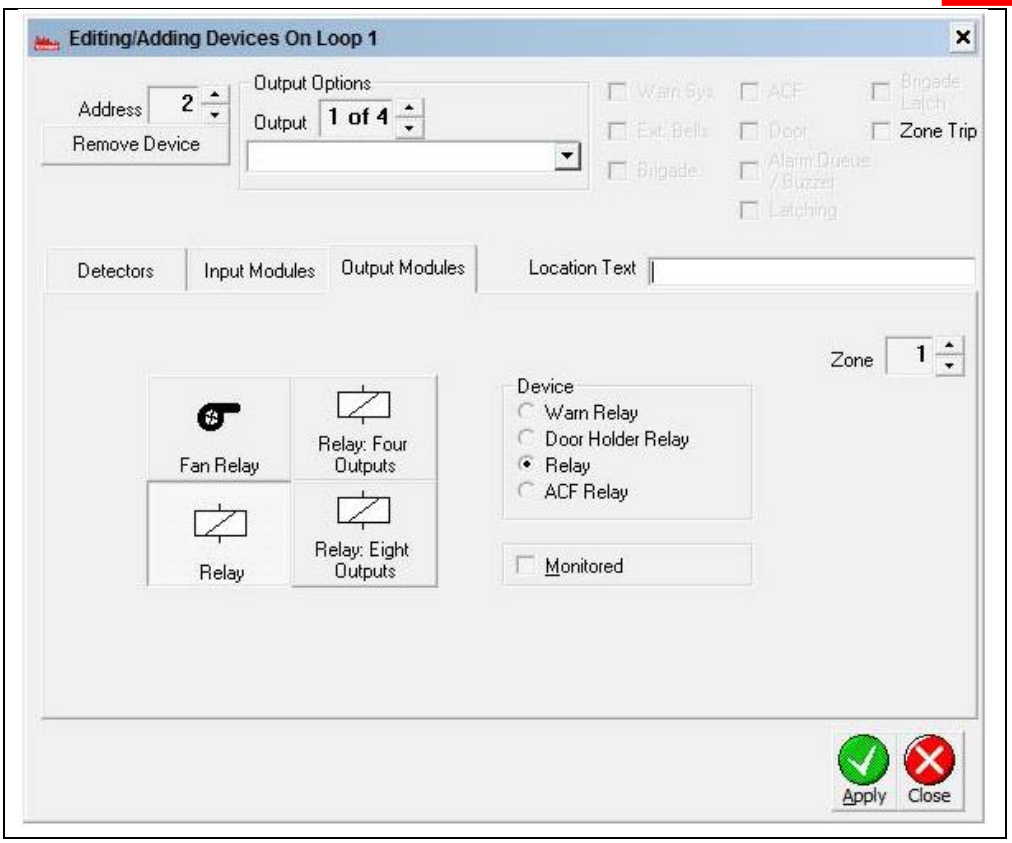

Figure 3.2: F100-128 Isolate Timer Module
Address 'n+1' = 2

3.2 F100A-256 Analogue Addressable Panel (s/w 5.xx)

3.2.1 Isolate Timer Module Configuration:

Key: Y = Yes N = No o = optional z = zone to be isolated	**Address**	**Type**	**Zone**	**Sens (Day)**	**Sens (Night)**	**Latching**	**AVF**	**Warn. Sys.**	**Ext Bell**	**ACF**	**Door**	**Fire**	**GP Relay**	**Fire Queue**	**Visible**	**Logged**	**Buzzer**	**Brigade Latch**	**Output 1**	**Output 2 - 4**
	n	ISO	z			N	N	N	N	N	N	N	N	N	Y	Y	N	N	o	o
	n+1	RLY	0																	

Table 3-2: Isolate Timer Module F100-256 Configuration

The 'Visible' and 'Logged' flags for the ITM device address should be set to YES; all other flags remain un-configurable. In 'Zones,' map the respective Zone's 'Isolated Alarm' output to the ITM RLY device. This will activate the Alarm Indication on the ITM LCD by displaying 'A' (for alarm) when the isolated zone is in alarm. In the Zone menu for the respective zone, map from output 'Isolated Alarm' to the ITM's relay address as shown below.

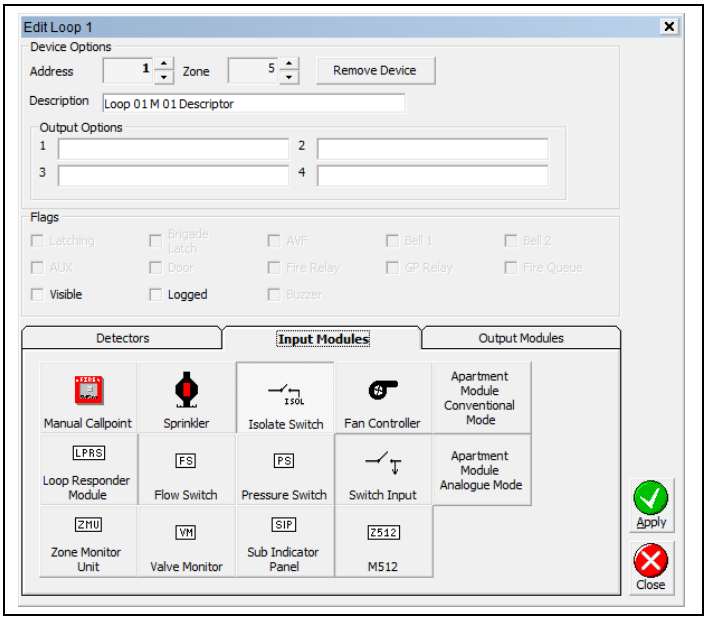

Figure 3.3: F100-256 Isolate Timer Module
Address 'n' = 1

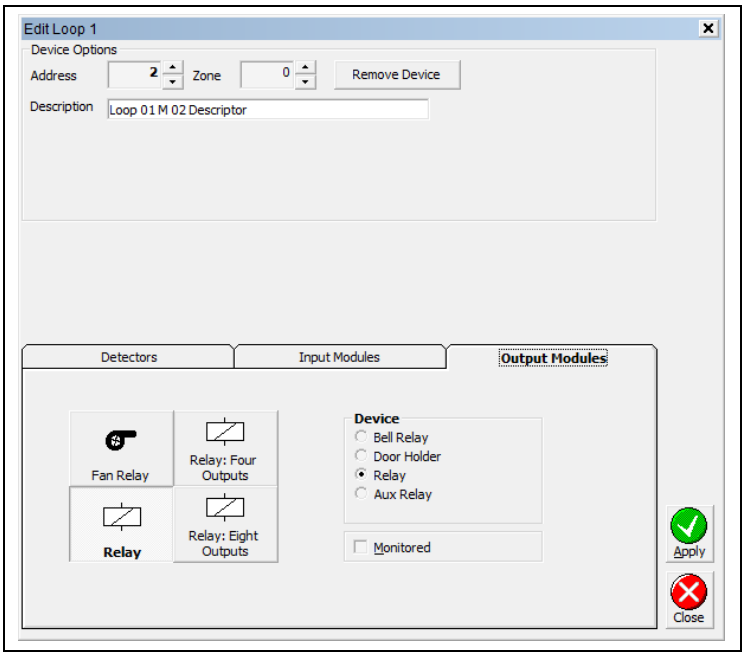

Figure 3.4: F100-256 Isolate Timer Module Relay
Address 'n' +1 = 2

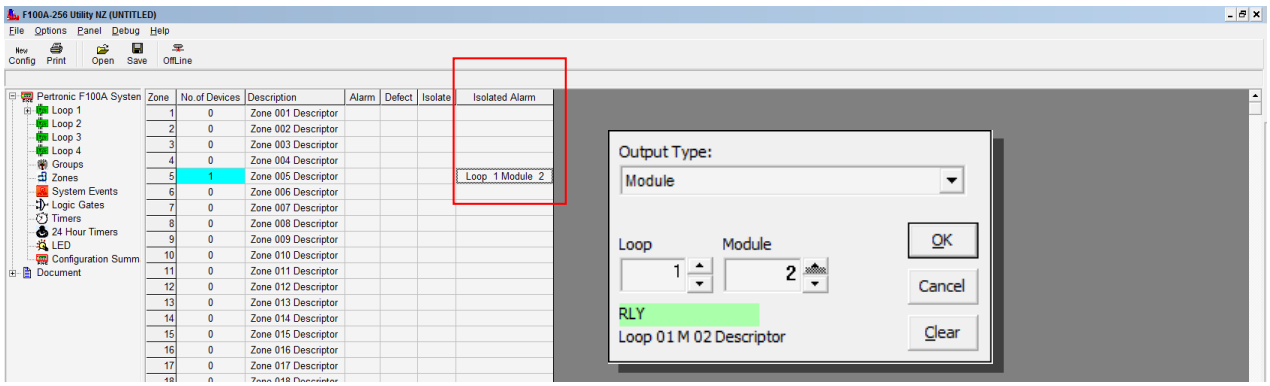

Figure 3.5: F100-256 Isolate Timer Module Relay
Address 'n' +1 = 2 mapped for Zone 5
Zone 5 output 'Isolated Alarm' mapped to Relay address.

3.3 F120A Analogue Addressable Panel

3.3.1 F120 Isolate Timer Module Configuration:

Key: Y = Yes N = No o = optional - = leave Blank z = zone to be isolated	Address	Type	Zone	Warn. Sys.	Ext. Bell	Brigade	ACF	Door	Alarm / Buzzer Queue	Latching	Brigade Latch	Zone Trip	AVF	Sens	Output 1	Output 2 - 4
	n	ISO	z	N	N	N	N	N	N	N	N	N	-	-	Pre-Alarm > ITMn+1 (RLY)	o
	n+1	RLY	0									**N**				

Table 3-3: Isolate Timer Module F120 Configuration

By default, all flags will remain un-configured, except for the RLY device's "Zone Trip" flag, which must be set to 'No'. Map an output of the ITM's ISO device to the ITM's RLY device, and set to activate on a 'Pre-Alarm'. This will activate when the zone ISO switch is active and there is an alarm condition in the mapped zone.

Note that the 'Pre-Alarm' event is simply used to instruct the F120 to operate in Zone Isolate Mode. If the 'Pre-Alarm' qualifier is not used, the output will trigger when the ISO switch is active, regardless of the alarm state of the zone.

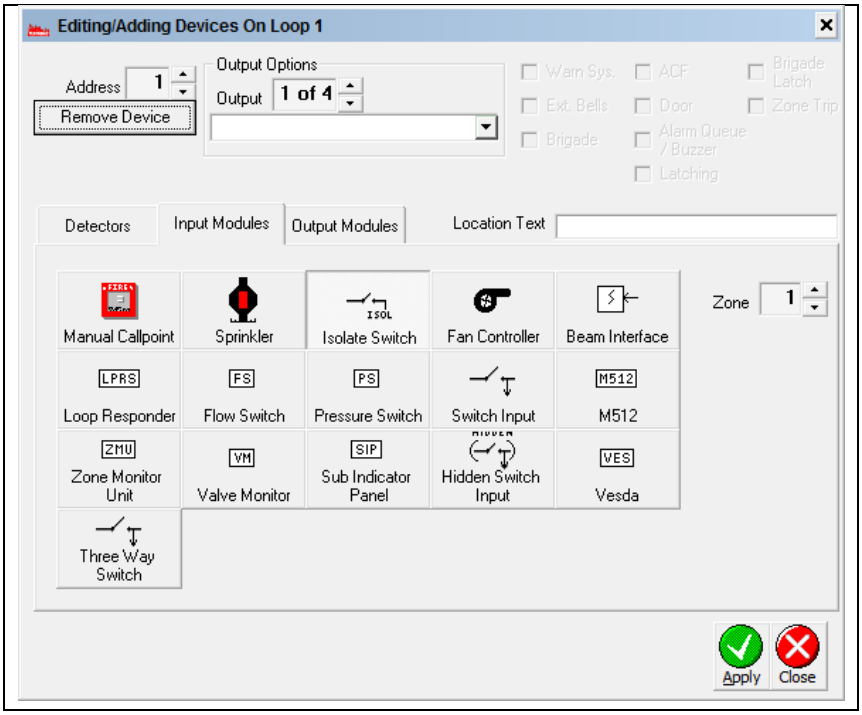

Figure 3.6: F120 Isolate Timer Module
Address 'n' = 1

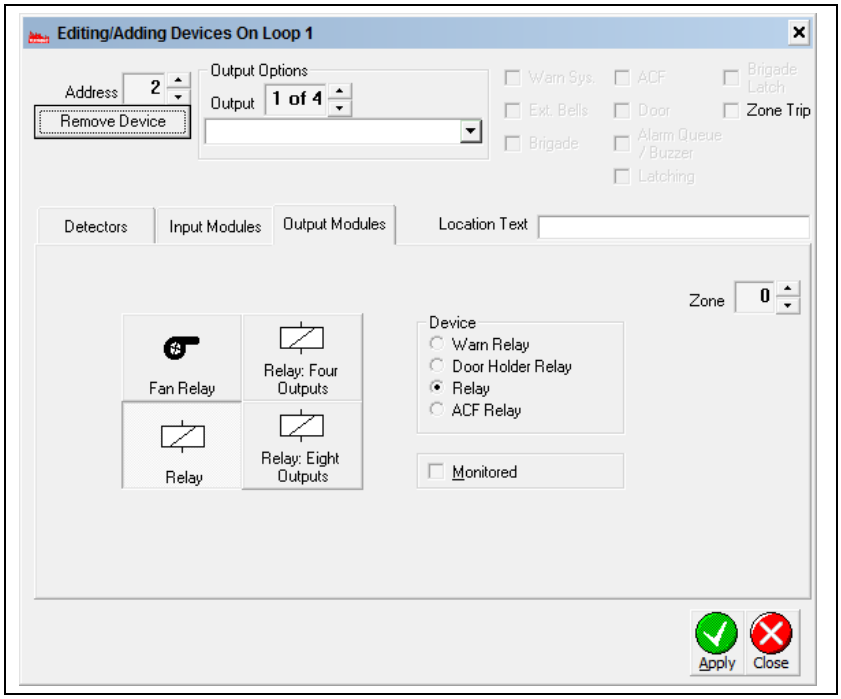

Figure 3.7: F120 Isolate Timer Module
Address ‘n+1’ = 2

3.4 F220 Analogue Addressable Panel

3.4.1 F220 Isolate Timer Module Configuration:

Key: Y = Yes/check N = No/no check o = optional - = leave Blank z = zone to be isolated	D M	Address	Location	Type	Sens (Day)	Zone	AAF/ADF	Bell 1	Bell 2	Brigade	Aux	Door	Alarm/Buzzer Queue	Latching	Brigade Latch	Zone Trip	AVF	Ancillary	Screen	History	Export	Outputs	Inputs	Outputs 1- 4
	M	n		ISO		z		N	N	N	N	N	N	N	N	N	-	-	-	-	-	1	0	-
	M	n+1		RLY		0		N	N	N	N	N	N	N	N	N	-	-	N	**Y**	N	0	0	-

Table 3-4: Isolate Timer Module F220 Configuration

In FireUtils, open the Loop Editor

- Select the ‘Switch Input (Disable)’ from the ‘Input Modules’ as shown in Figure 3-8 (Red square) by double clocking the left mouse button when the cursor is over the Type cell. Alternatively, select Tools > Device Selector > Input Modules > Switch Input [Disable] in the Menu Bar
- In this example, the module address (n) of the ‘Switch Input (Disable)’ has been set to 1 (Figure 3-8, Purple circle)
- Select ‘Relay’ from the ‘Output Modules’ (Figure 3-8, Green square) in the next module address (n+1, Purple circle) which, in this example, is 2. Alternatively select Tools > Device Type Selector > Output Modules > Relay in the Menu Bar.

By default, all the flags will be unchecked, except the Relay History. When the Zone is set for the ‘Switch Input’ to a non-zero number (1 in this example), the number in the ‘O/Ps’ cell (Figure 3-9, Blue circle) changes to 1. This changes to 2 when the next step is completed.

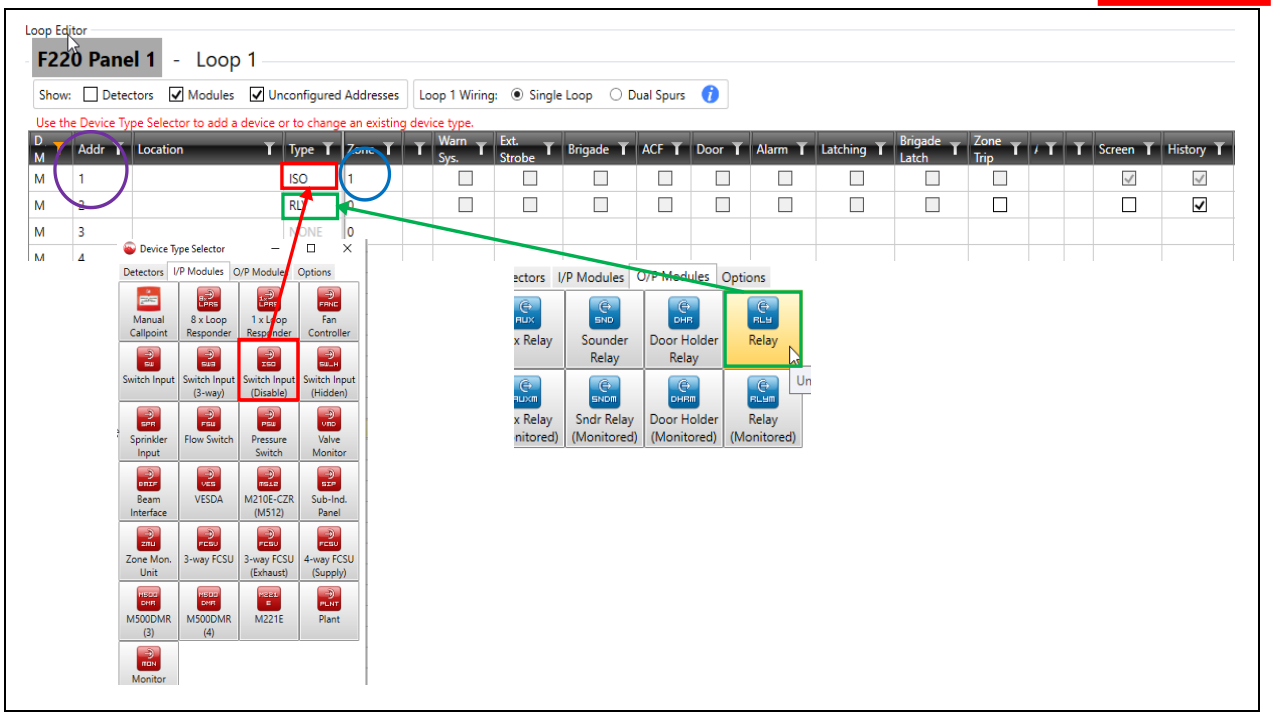

Figure 3-8: Configuring the ITM in FireUtils' Loop Editor

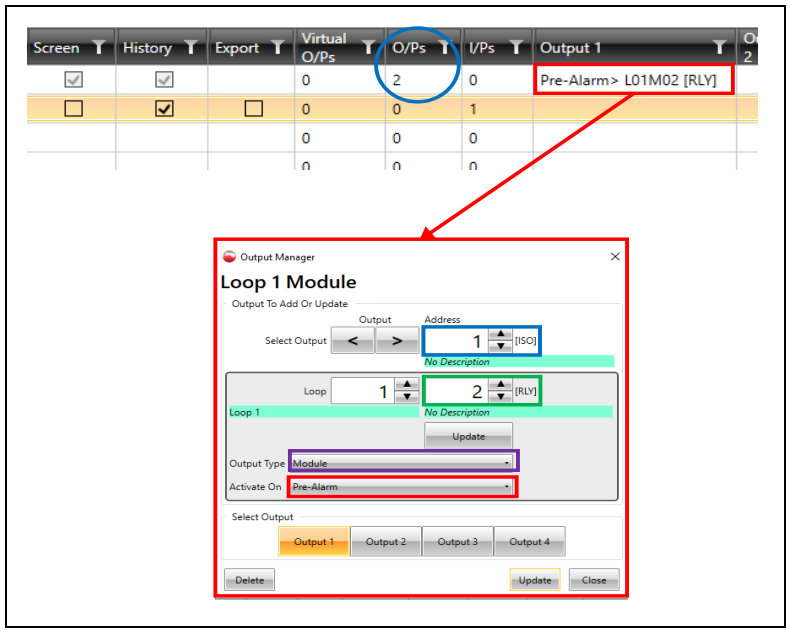

Figure 3-9: Mapping the Relay to the Isolate Module

In FireUtils' Loop Editor

- Position the mouse cursor over the blank 'Output 1' cell of the 'Switch Input (Disable)' and double click the left mouse button. This opens the Output Manager as shown in Figure 3-9.
- In 'Output Manager' the address of 'ISO' module will be 1 (Figure 3-9, Blue rectangle).
- Increment the address (Figure 3-9, Green rectangle) to 2. This is the address of the Relay.
- In 'Output Type', select Module from the drop down list (Figure 3-9, Purple rectangle).
- In 'Activate On', select Pre-Alarm from the drop down list (Figure 3-9, Red rectangle).
- Click on the 'Update' button and 'Close'.

The ITM's ISO device is now set to activate its RLY on a 'Pre-Alarm'. That is, it will activate when the zone ISO switch is active and there is a pre-alarm condition in the mapped zone. If the 'Pre-Alarm' qualifier is not used, the output would trigger every time the ISO switch was active, regardless of the alarm state of the zone.

4 ORDER CODES

Description	Product Code
Analogue Addressable Isolate Timer Module	ITM

Table 4-1: Product Codes

Document Change History

Issue Number	Date	Reason for Update	Description of Changes	Author
Issue 1:	13 November	Initial release.		ShaneR
Issue 2:	23 Sept 2010	Clarify wording and update screen views Specify differences between F100A-128 and F100A-256 configurations	Clarify configuration wording and update screen views on pages 9, 10, and 11. Screen view changes and clarifications made to Section 3, Fire Panel Configuration	JeremyE PeterF
Issue 2.1	October 2012	F100A versions not clear	Section 1.2 Added F100A-128& 256 version info	RDB
Issue 2.2	Jan 2014	CR 2069, 2070	Clarify Isolation precedence	RDB
Issue 2.3	Dec 2020	CN2906	Timer accuracy, F220 configuration	RJK
Issue 2.4	Oct 2024	CN3587	Reformatted document. Updated F220 configuration	RJK

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